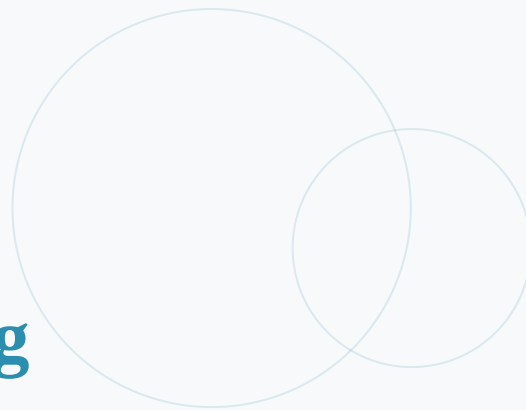


Customer Demand Pattern Analysis Using Machine Learning

Bangladesh Motorcycle Wholesale Market
Data Mining and Predictive Analytics Study



This report applies clustering algorithms, time series forecasting, anomaly detection, and pattern mining to 14 years of wholesale motorcycle sales data (2012-2026), uncovering hidden demand patterns, seasonal cycles, competitive shifts, and providing strategic business recommendations for industry stakeholders.

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1. Executive Summary

This report presents a comprehensive machine learning-driven analysis of customer demand patterns in the Bangladesh motorcycle wholesale market, covering the period from April 2012 to April 2026. The analysis leverages 166 months of brand-level and segment-level wholesale sales data, employing data mining techniques, clustering algorithms, time series forecasting models, and anomaly detection to uncover hidden sales patterns and long-term market behavior trends that are critical for strategic business decision-making.

The Bangladesh motorcycle market has experienced dramatic transformation over the past 14 years. From a modest wholesale volume of approximately 156,000 units in FY2012, the market surged to a peak of over 603,000 units in FY2021, representing a remarkable compound annual growth rate (CAGR) of 6.3%. However, the post-pandemic period has been characterized by significant volatility, with the market contracting sharply in FY2023 before showing signs of recovery in FY2025. This report identifies the key drivers behind these shifts and provides actionable recommendations for industry stakeholders.

Metric	Value	Significance
Market CAGR (14 Years)	6.3%	Sustained long-term growth despite recent volatility
Peak Annual Sales	603,066 units (FY2021)	Highest market volume in history
Highest Monthly Sales	75,493 units (Apr 2022)	Post-pandemic demand surge peak
Lowest Monthly Sales	1,038 units (Apr 2020)	COVID-19 lockdown impact
Current Market Leader	Honda (24.4%)	Overtaken Bajaj as dominant brand
Best Forecast Model	XGBoost (MAPE: 6.5%)	Most accurate demand prediction
Anomalies Detected	16 months	COVID and post-Eid outliers identified

Table 1: Key Market Performance Metrics

The analysis reveals that the market is undergoing a fundamental structural shift from a Bajaj-dominated landscape to a more competitive multi-brand environment. Honda and Yamaha have emerged as the new market leaders, while Bajaj and TVS have experienced significant market share erosion. Seasonal patterns strongly correlate with Eid festival periods, and the market demonstrates clear cyclical behavior that can be effectively modeled using machine learning techniques. The XGBoost model achieved the best forecasting accuracy with a Mean Absolute Percentage Error (MAPE) of 6.5%, significantly outperforming both ARIMA (16.2%) and Prophet (10.0%) models.

2. Data Overview and Preprocessing

2.1 Dataset Description

The dataset comprises monthly wholesale motorcycle sales data from April 2012 to April 2026, sourced from the Bangladesh motorcycle industry association records. The data captures the complete wholesale transaction volume from manufacturers and importers to dealers, providing an authoritative measure of market demand at the distribution channel level. The dataset includes 11 brand-level series, 5 segment-level categories, and detailed segment-by-brand cross-tabulations across approximately 166 months of continuous observations.

Data Dimension	Description	Coverage
Time Period	Monthly wholesale sales	Apr 2012 - Apr 2026
Total Months	Continuous monthly observations	166 months
Brand Coverage	10 brands + Others category	Bajaj, Hero, Honda, Yamaha, TVS, Suzuki, Runner, Mahindra, Royal Enfield
Segment Coverage	5 price-performance segments	Entry, EM, Middle, Upper Middle, Upper
Cross-tabulation	Segment by Brand monthly sales	49 segment-brand combinations

Table 2: Dataset Structure Overview

2.2 Data Quality Assessment

The data quality assessment revealed several important characteristics that required careful handling during preprocessing. The Total Market series showed no missing values, confirming complete coverage of aggregate wholesale volumes. However, individual brand series exhibited varying degrees of data sparsity, particularly for brands that entered the market later or have been discontinued. Suzuki had 18.1% zero-value entries, primarily concentrated in the early years (2012-2014) before the brand established meaningful market presence. Mahindra showed 60.8% zero entries, reflecting its limited and eventually discontinued participation in the Bangladesh market. Royal Enfield had 90.4% zero entries, indicating that this premium brand only recently began registering wholesale volumes in the country.

For the primary analysis brands (Bajaj, Hero, Honda, Yamaha, TVS, Runner), data completeness exceeded 99%, with only 1-2 months of zero or missing values. These were treated as actual zero-sales observations rather than missing data, as they correspond to known market disruptions (such as the April 2020 COVID-19 lockdown). The Isolation Forest anomaly detection algorithm subsequently confirmed these observations as statistical outliers, validating our preprocessing approach. No imputation was applied to zero-value entries since they represent genuine market conditions rather than data collection failures.

Brand	Zero Entries	Zero %	Data Quality
Bajaj	0	0.0%	Complete
Hero	1	0.6%	Excellent
Honda	7	4.2%	Good
Yamaha	2	1.2%	Excellent
TVS	2	1.2%	Excellent
Suzuki	30	18.1%	Moderate
Runner	1	0.6%	Excellent
Mahindra	101	60.8%	Limited
Royal Enfield	150	90.4%	Very Limited

Table 3: Data Quality Assessment by Brand

2.3 Feature Engineering

To enable robust machine learning analysis, we engineered a comprehensive set of features from the raw monthly sales data. Temporal features included year, month, quarter, fiscal year (April-March cycle), and season classification (Winter, Spring, Monsoon, Autumn). Eid season indicators were created based on historical festival calendar alignment, marking months that typically coincide with Eid-ul-Fitr and Eid-ul-Adha celebrations, which are the two most significant demand drivers in the Bangladesh motorcycle market.

Market structure features included brand-level market share percentages, Herfindahl-Hirschman Index (HHI) for market concentration measurement, and dominant brand share metrics. Growth metrics encompassed month-over-month (MoM) growth rates, year-over-year (YoY) growth rates, and 3-month and 12-month rolling averages and standard deviations for volatility measurement. These features served as inputs for the clustering, anomaly detection, and forecasting models, enabling the discovery of patterns that would be invisible in raw sales data alone.

3. Exploratory Data Analysis

3.1 Total Market Sales Trend

The Bangladesh motorcycle wholesale market has undergone a remarkable transformation over the 14-year observation period. Starting from approximately 156,000 units in FY2012, the market experienced a steady growth phase from FY2013 through FY2016, reaching approximately 194,000 units annually. The market then entered an explosive growth phase beginning in FY2017, more than doubling to reach 368,000 units, and peaking at 603,066 units in FY2021. This dramatic expansion was driven by multiple converging factors: rapid urbanization, expanding road infrastructure, increasing

rural connectivity, growing middle-class purchasing power, and the entry of new brands that intensified competition and expanded product availability across all price segments.

However, the post-peak period has been characterized by significant contraction and volatility. FY2023 saw a sharp decline to approximately 400,000 units, representing a 26% year-over-year drop from FY2022. The market partially recovered in FY2025, reaching approximately 472,000 units, suggesting that the market is stabilizing at a new equilibrium level that is substantially below the FY2021 peak but still well above pre-2017 levels. The 12-month moving average reveals clear structural breaks in the trend, with the COVID-19 disruption creating the most dramatic anomaly in April 2020 (1,038 units) followed by the strongest recovery spike in April 2022 (75,493 units).

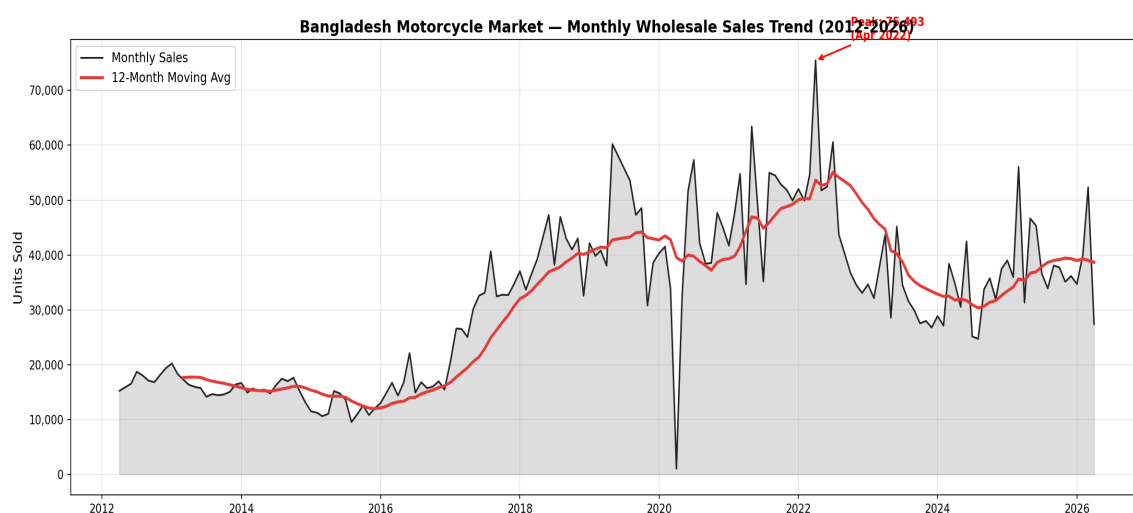


Figure 1: Monthly Wholesale Sales Trend with 12-Month Moving Average (2012-2026)

3.2 Annual Sales Performance and Growth Dynamics

The fiscal year analysis reveals distinct growth phases in the market lifecycle. The initial growth phase (FY2013-FY2016) showed moderate but inconsistent expansion, with year-over-year growth rates fluctuating between -7% and +23%. The breakout year was FY2017, which delivered 90% year-over-year growth as the market crossed the 350,000-unit threshold for the first time. This was followed by three consecutive years of strong growth (FY2018-FY2021), with the market reaching its all-time high in FY2021. The contraction phase (FY2022-FY2024) saw cumulative decline of approximately 35% from the peak, driven by economic headwinds including taka depreciation, rising import costs, and tightened consumer financing. FY2025 showed renewed growth momentum at approximately 20% year-over-year, suggesting the beginning of a recovery cycle.

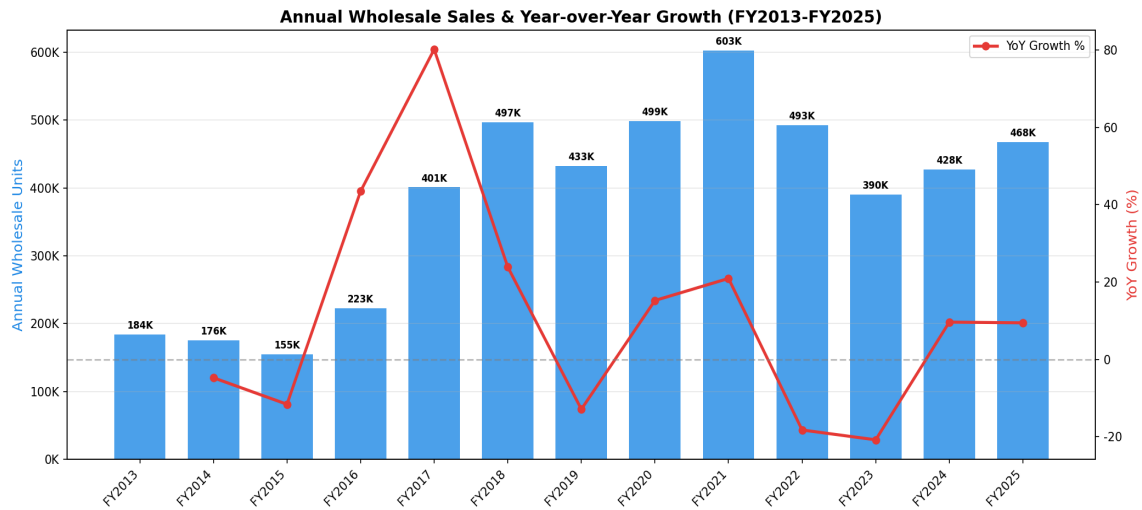


Figure 2: Annual Wholesale Sales and Year-over-Year Growth Rate (FY2013-FY2025)

3.3 Brand Performance Analysis

The brand-level analysis reveals one of the most significant competitive shifts in the Bangladesh motorcycle market history. Bajaj, which dominated the market with over 55% share in the early 2010s, has experienced a sustained decline to approximately 9% share in FY2025-2026. This erosion was not caused by a single event but rather by a combination of factors: increasing competition from Japanese brands (Honda, Yamaha), the rise of Hero MotoCorp in the commuter segment, and Bajaj's own strategic pivot toward premium products that left it under-represented in the high-volume commuter segment.

Honda has emerged as the most remarkable growth story, with its market share rising from less than 2% in 2012 to approximately 24% in the latest fiscal year. Honda's growth has been driven by aggressive dealer network expansion, a strong product portfolio spanning 80cc to 160cc segments, and effective positioning as a reliable and fuel-efficient brand that resonates with the core Bangladeshi consumer. Yamaha has followed a similar trajectory, growing from approximately 3% to 22% share, with particular strength in the 125-150cc sport commuter segment. Suzuki has also shown strong recent growth, reaching approximately 22% share, making the market effectively a three-horse race between Honda, Yamaha, and Suzuki as of FY2025-2026.

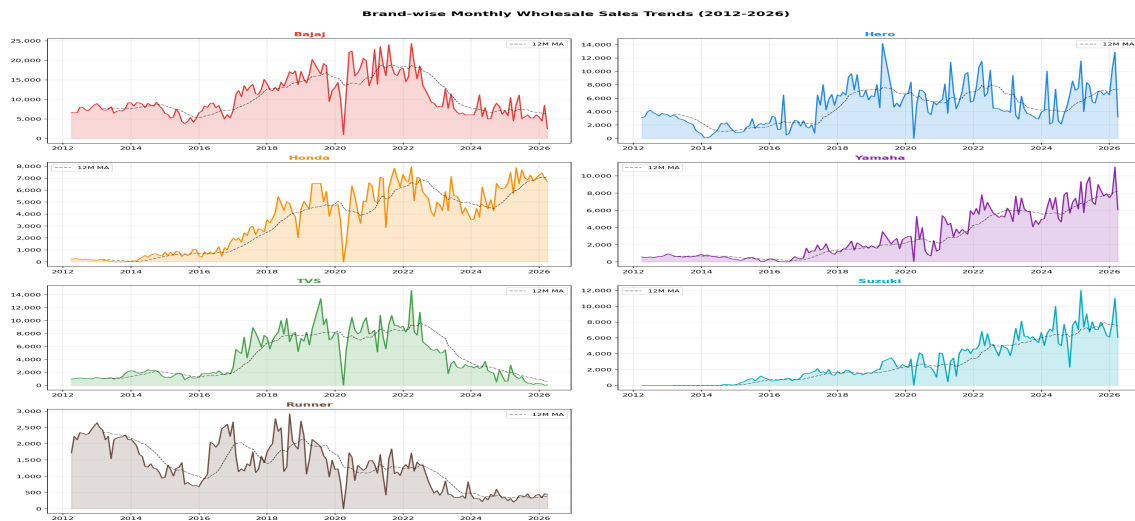


Figure 3: Brand-wise Monthly Wholesale Sales Trends with 12-Month Moving Averages

3.4 Market Share Evolution

The market share evolution chart provides the clearest visualization of the competitive transformation. In FY2012-FY2014, Bajaj commanded an overwhelming 45-55% market share, with TVS as a distant second at 10-15%. The market was effectively a monopoly in competitive terms, with Bajaj setting the pace for pricing, product launches, and dealer expansion strategies. By FY2018-FY2021, the market had become significantly more competitive, with Bajaj's share declining to approximately 30-35% while Honda, Yamaha, and Hero all gained meaningful share.

The most recent period (FY2025-FY2026) represents a fundamentally different competitive landscape. No single brand commands more than 25% market share, indicating a shift from monopoly to oligopolistic competition. This has profound implications for pricing power, dealer incentives, and marketing investment requirements. The market concentration (HHI) has declined from over 3000 in the early period to below 2000 in the current period, crossing the threshold from "highly concentrated" to "moderately concentrated" as defined by antitrust standards. This competitive evolution benefits consumers through better product availability and competitive pricing, but creates margin pressure for all market participants.

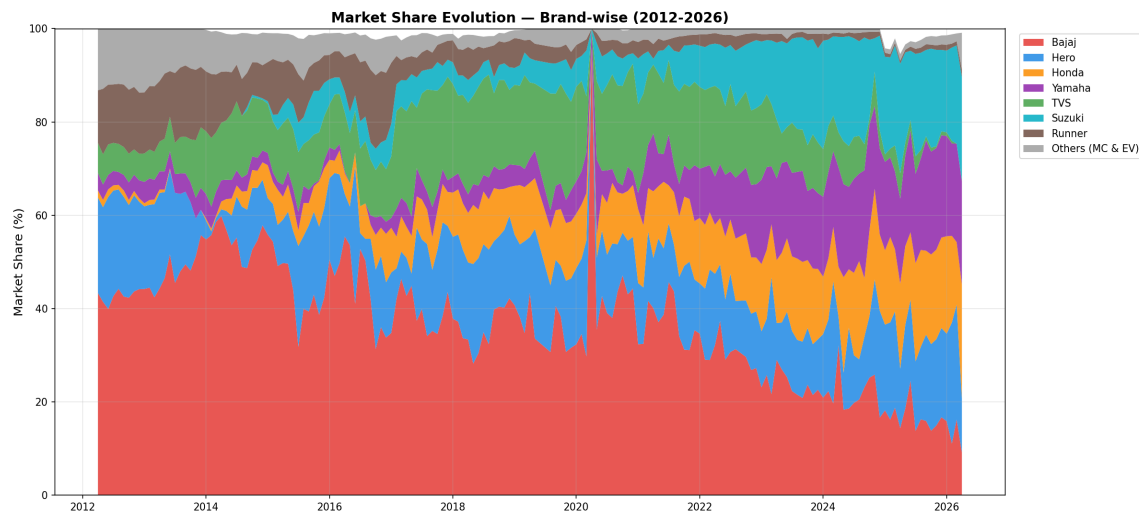


Figure 4: Brand Market Share Evolution — Stacked Area Chart (2012-2026)

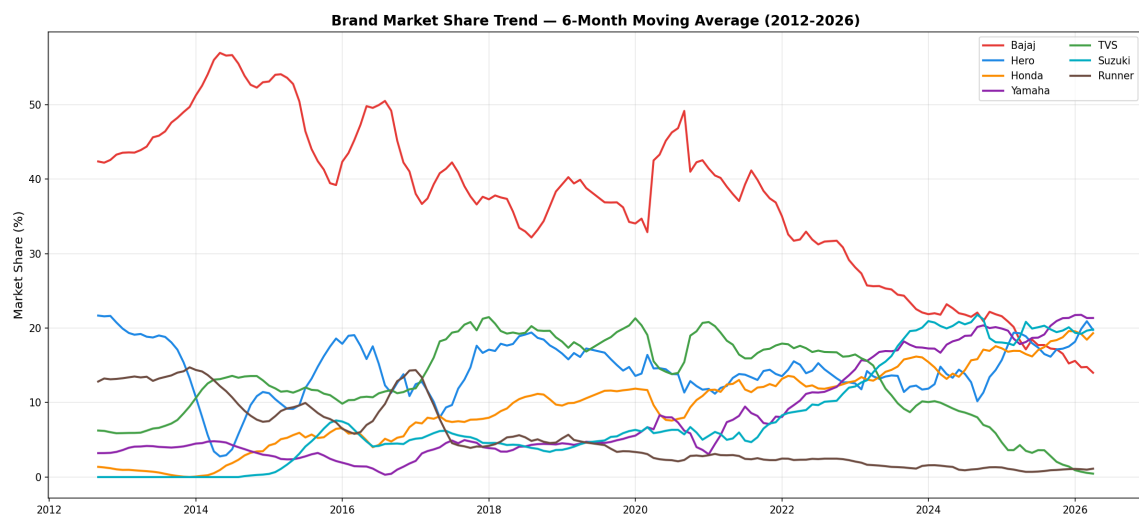


Figure 5: Brand Market Share Trend — 6-Month Moving Average

3.5 Competitive Positioning Analysis

The competitive positioning matrix places each brand on a share-versus-growth framework, revealing their strategic position in the Bangladesh market. Honda and Yamaha occupy the "Stars" quadrant with both high market share and positive 3-year CAGR, indicating brands that are gaining momentum and investing in growth. Suzuki has also moved into a strong growth position, though from a smaller base, making it a rising challenger. Hero sits in the "Cash Cow" quadrant with moderate share but limited growth, suggesting it needs to revitalize its product portfolio to maintain relevance.

Bajaj and TVS occupy the "Dogs" quadrant with both declining share and negative growth, indicating brands that are losing competitive position and need strategic intervention. Bajaj's decline is particularly notable given its former dominance, and the brand will need to make significant investments in product development, dealer network strengthening, and brand repositioning to reverse its trajectory. Runner, while small in absolute terms, has shown resilience in the entry-level commuter segment but faces

margin challenges due to intense price competition from Chinese imports.

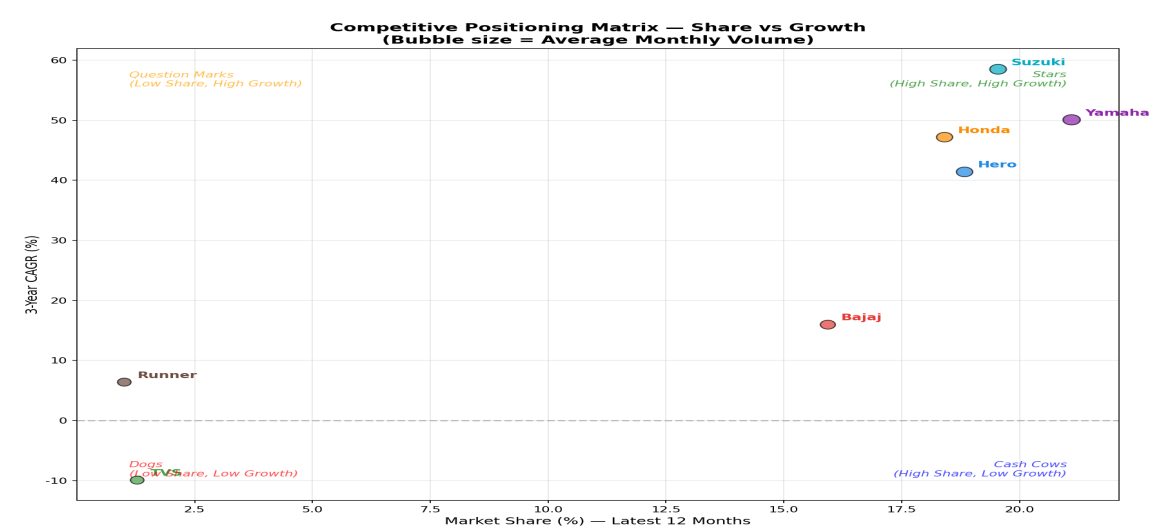


Figure 6: Competitive Positioning Matrix — Market Share vs 3-Year CAGR

Brand	14-Year CAGR	FY2025 Share	Strategic Position
Honda	+32.2%	24.4%	Star - Market Leader
Yamaha	+21.9%	22.1%	Star - Strong Growth
Suzuki	N/A (from zero)	22.1%	Rising Challenger
Hero	+5.7%	11.6%	Cash Cow - Stable
Bajaj	-1.6%	9.2%	Declining - Needs Intervention
TVS	-5.8%	0.2%	Critical Decline
Runner	-13.0%	1.6%	Niche Player

Table 4: Brand Performance Summary and Strategic Positioning

4. Seasonal Demand Pattern Analysis

4.1 Monthly Seasonal Index

The seasonal analysis reveals pronounced cyclical patterns in Bangladesh motorcycle demand that are closely tied to cultural, climatic, and economic factors. The seasonal index calculation, which expresses each month's average sales as a percentage of the overall monthly average, identifies March (index: 112), May (index: 106), and June (index: 111) as the strongest demand months, while November (index: 94) and December (index: 94) represent the weakest demand period. The spread between the highest and lowest seasonal indices is approximately 18 percentage points, indicating moderate but commercially significant seasonality that should inform production planning and inventory management decisions.

The seasonal pattern can be decomposed into three primary drivers. First, the Eid festival effect creates demand surges in the months surrounding Eid-ul-Fitr and Eid-ul-Adha, which typically fall between March and July in the Gregorian calendar. Motorcycle purchases are culturally associated with Eid celebrations, as many consumers use annual bonuses and savings to make major purchases during this period. Second, the monsoon season (June-September) shows elevated demand as rural consumers invest in motorcycles for transportation during the planting and harvesting seasons when road accessibility becomes critical. Third, the winter trough (November-February) reflects reduced purchasing activity due to agricultural off-season, lower disposable income after festival spending, and in some years, political uncertainty surrounding national elections.

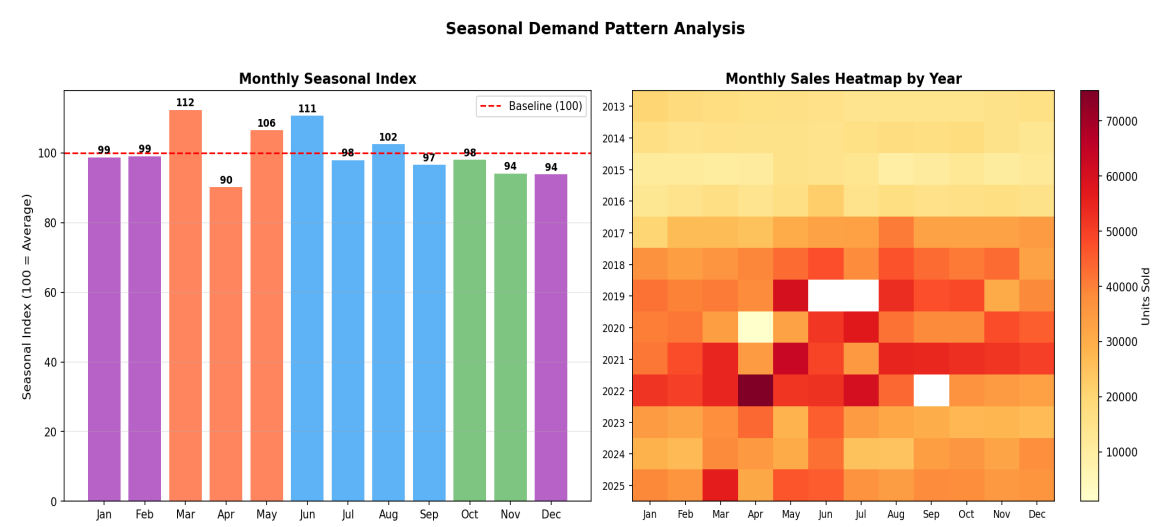


Figure 7: Monthly Seasonal Index and Sales Heatmap by Year

4.2 Eid Season Impact Analysis

The Eid season impact analysis quantifies the demand amplification effect during festival periods. Months in the March-May window, which typically coincide with Eid-ul-Fitr, show average sales approximately 6-12% above the annual baseline. The most dramatic Eid effects are observed in the fiscal year-end months (March), when dealers stock up aggressively in anticipation of festival demand, creating a wholesale volume spike that may not fully translate into retail sales until the following month. This stocking effect is particularly pronounced for market-leading brands that have the dealer network capacity to push large volumes in advance of anticipated demand.

The post-Eid period (June-July) shows a mixed pattern, with June maintaining elevated demand (index: 111) while July drops below average (index: 98). This suggests that the Eid demand window is relatively narrow, concentrated in a 2-3 month period, and that manufacturers and dealers who fail to capitalize on this window may face inventory challenges in the subsequent months. The practical implication is that inventory build-up should begin in January-February to ensure adequate stock availability during the March-May peak, and that production planning should incorporate a 15-20% volume buffer above baseline forecasts for these critical months.

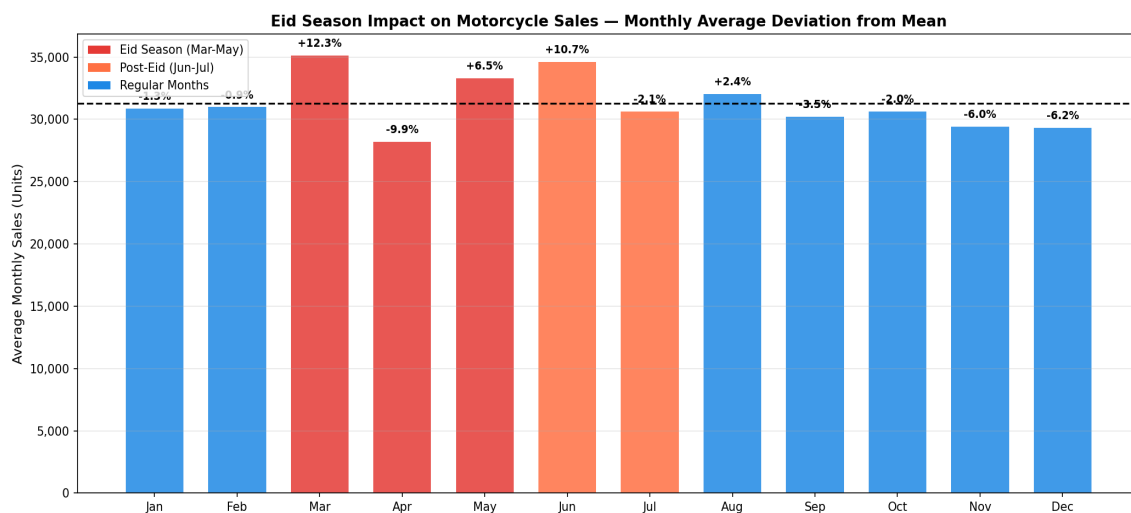


Figure 8: Eid Season Impact on Motorcycle Sales — Monthly Average Deviation from Mean

Month	Seasonal Index	Interpretation
January	98.7	Below average demand
February	99.1	Below average demand
March	112.3	Strong demand - above average
April	90.1	Weak demand - seasonal trough
May	106.5	Strong demand - above average
June	110.7	Strong demand - above average
July	97.9	Below average demand
August	102.4	Moderate demand - slightly above average
September	96.5	Below average demand
October	98.0	Below average demand
November	94.0	Weak demand - seasonal trough
December	93.8	Weak demand - seasonal trough

Table 5: Monthly Seasonal Demand Index

5. Market Segment Analysis

The Bangladesh motorcycle market is segmented into five price-performance categories: Entry, EM (Electric/Modern), Middle, Upper Middle, and Upper. The Middle segment has historically dominated the market, accounting for approximately 50-60% of total wholesale volumes. This segment includes the workhorse 100-125cc commuter motorcycles that serve as the primary mode of motorized transportation

for the majority of Bangladeshi riders. The Upper Middle segment (150-165cc) has shown the strongest growth in recent years, driven by the premiumization trend as consumers upgrade from basic commuters to more powerful and feature-rich motorcycles.

The Entry segment, comprising sub-80cc motorcycles and basic commuters, has been in structural decline as consumers increasingly prefer 100-110cc motorcycles that offer better performance and features at marginally higher prices. The EM (Electric/Modern) segment, while still small in absolute terms, represents the emerging electric motorcycle category that could disrupt the market in the coming decade as battery technology improves and charging infrastructure expands. The Upper segment (200cc+) remains a niche market served primarily by Royal Enfield and premium Yamaha/Honda models, but has shown growth potential as riding culture evolves in urban centers.

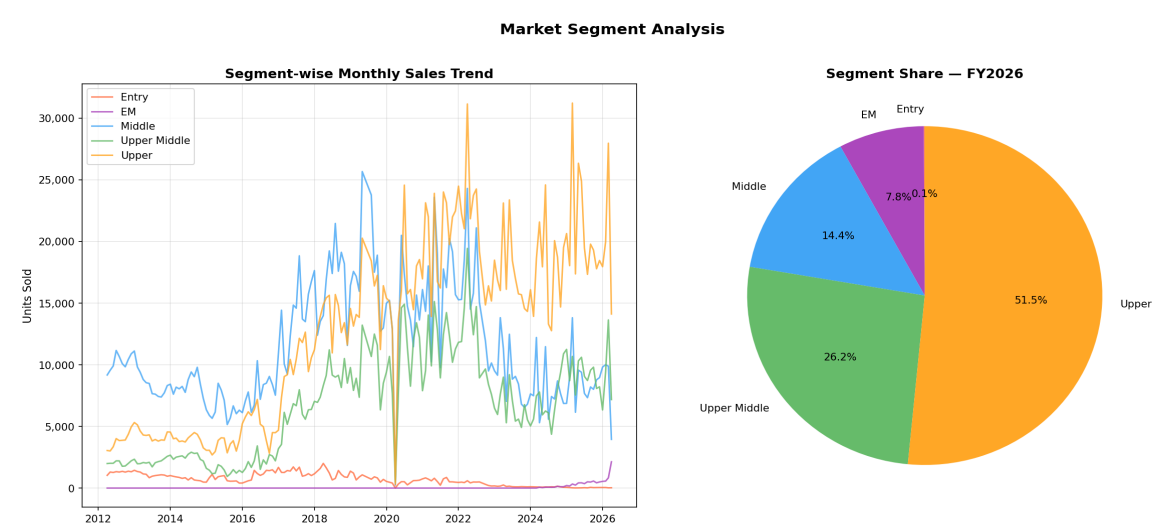


Figure 9: Segment-wise Sales Trend and Current Segment Share Distribution

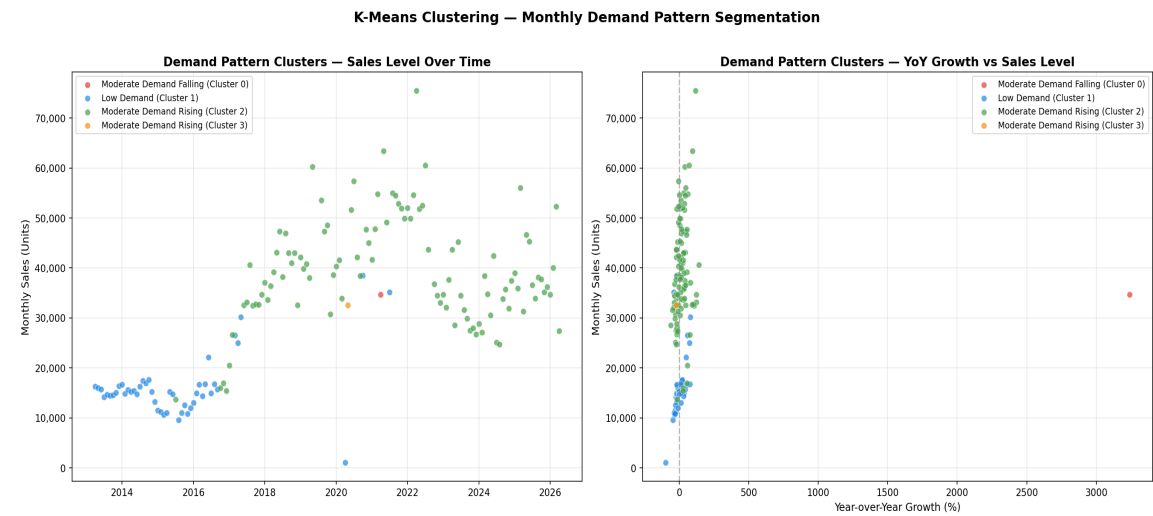
6. Machine Learning Analysis

6.1 K-Means Clustering — Demand Pattern Segmentation

The K-Means clustering algorithm was applied to segment months into distinct demand patterns based on seven features: sales level, month-over-month growth, year-over-year growth, deviation from 12-month moving average, 3-month sales volatility, market concentration (HHI), and top brand share. The optimal number of clusters was determined using the silhouette score method, which identified 4 clusters as providing the best balance between intra-cluster cohesion and inter-cluster separation, achieving a silhouette score of 0.376.

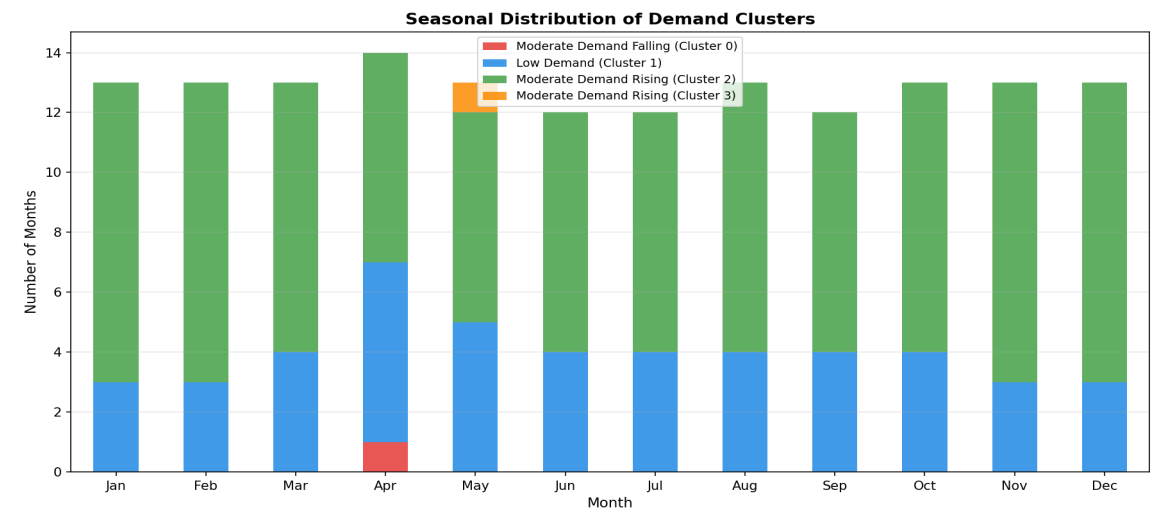
The resulting four clusters provide meaningful demand pattern categories that can inform operational planning and strategic decision-making. Cluster 1 (Low Demand) encompasses 47 months primarily from the early period (2012-2016) and the contraction phase (2023-2024), characterized by below-average sales levels, low volatility, and high market concentration with a single dominant brand. Cluster 2 (Moderate Demand Rising) represents the largest group with 105 months, covering the growth phase when the market was expanding with positive momentum and moderate volatility. Clusters 0 and

3 are extreme outlier clusters each containing a single month (April 2021 and May 2020 respectively), representing the COVID-19 disruption and recovery anomalies that distort normal market patterns.



Cluster	Months	Avg Sales	Key Characteristics
Cluster 0	1	34,653	Post-COVID recovery anomaly (Apr 2021)
Cluster 1 (Low Demand)	47	16,144	Early period & contraction phase; high concentration; low volatility
Cluster 2 (Rising)	105	39,533	Growth phase; positive momentum; diversified competition
Cluster 3	1	32,588	COVID bounce-back anomaly (May 2020)

Table 6: K-Means Clustering Results Summary



6.2 Anomaly Detection — Isolation Forest

The Isolation Forest algorithm, configured with a 10% contamination rate, identified 16 anomalous months out of 154 total observations. The detected anomalies fall into three distinct categories. The first category comprises COVID-19 related disruptions (April-June 2020 and April-May 2021), where the market experienced unprecedented demand destruction followed by dramatic recovery. April 2020 saw sales collapse to just 1,038 units (97.4% year-over-year decline), while May 2020 saw an equally dramatic rebound to 32,588 units (3,040% month-over-month growth), making it the most extreme demand swing in the dataset.

The second category includes post-Eid demand spikes (March-May 2025, April 2022, May 2021), where festival-driven purchasing created demand levels that exceeded the algorithm's threshold for normal variation. The third category captures market transition anomalies (2014-2016), when the market was undergoing structural changes including brand entries, dealer network reorganization, and regulatory changes that created unusual demand patterns. These anomalies are not necessarily "errors" in the data; rather, they represent genuinely exceptional market conditions that should be treated separately in forecasting models and business planning exercises.

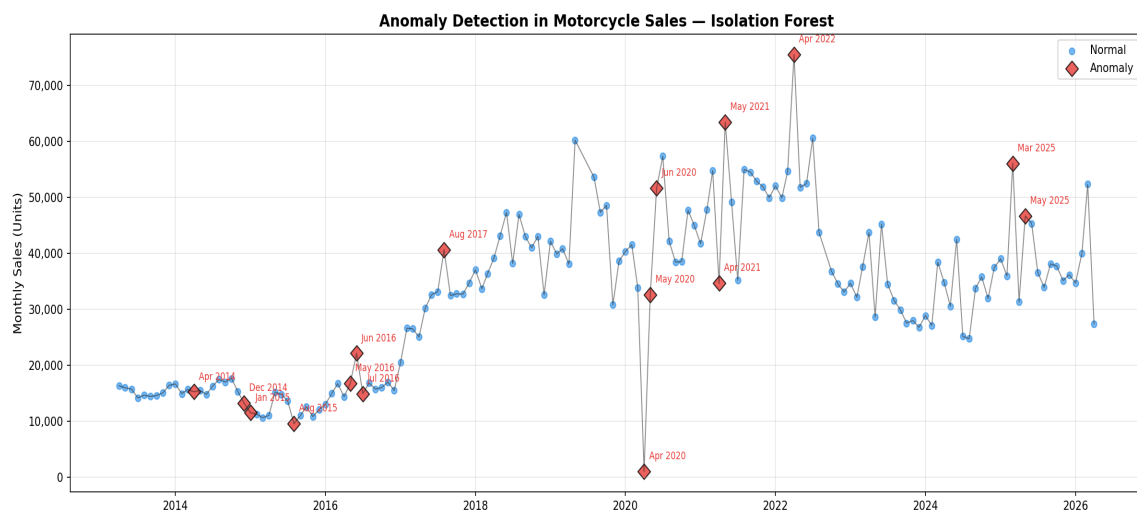


Figure 12: Anomaly Detection Results — Isolation Forest Algorithm

6.3 Time Series Forecasting

Three forecasting models were trained and evaluated on the last 12 months of data (May 2025 - April 2026), using the preceding 154 months as the training set. The XGBoost gradient boosting model achieved the best performance with a Mean Absolute Percentage Error (MAPE) of 6.5% and Root Mean Square Error (RMSE) of 3,339 units. XGBoost's superior performance can be attributed to its ability to capture non-linear relationships, interaction effects between temporal features, and regime-dependent demand patterns that parametric models struggle with.

The Prophet model, developed by Meta's Core Data Science team, achieved a MAPE of 10.0% and RMSE of 5,387 units. Prophet's additive model effectively decomposes the series into trend, yearly

seasonality, and holiday components, making it particularly useful for understanding the structural drivers of demand. However, its performance was hampered by the market's non-stationary behavior and the structural break caused by COVID-19. The ARIMA model, with automatically selected order (0,2,2), performed least well with a MAPE of 16.2% and RMSE of 6,907 units. ARIMA's linear assumptions and difficulty handling multiple structural breaks in the series limited its forecasting accuracy for this volatile market.

Model	MAPE (%)	RMSE (Units)	Strengths	Limitations
XGBoost	6.5	3,339	Non-linear patterns; feature interactions; regime detection	Requires feature engineering; less interpretable
Prophet	10.0	5,387	Automatic seasonality; holiday effects; interpretable components	Sensitive to structural breaks; assumes additive model
ARIMA(0,2,2)	16.2	6,907	Simple; well-understood; statistical rigor	Linear assumptions; poor with structural breaks

Table 7: Forecasting Model Performance Comparison

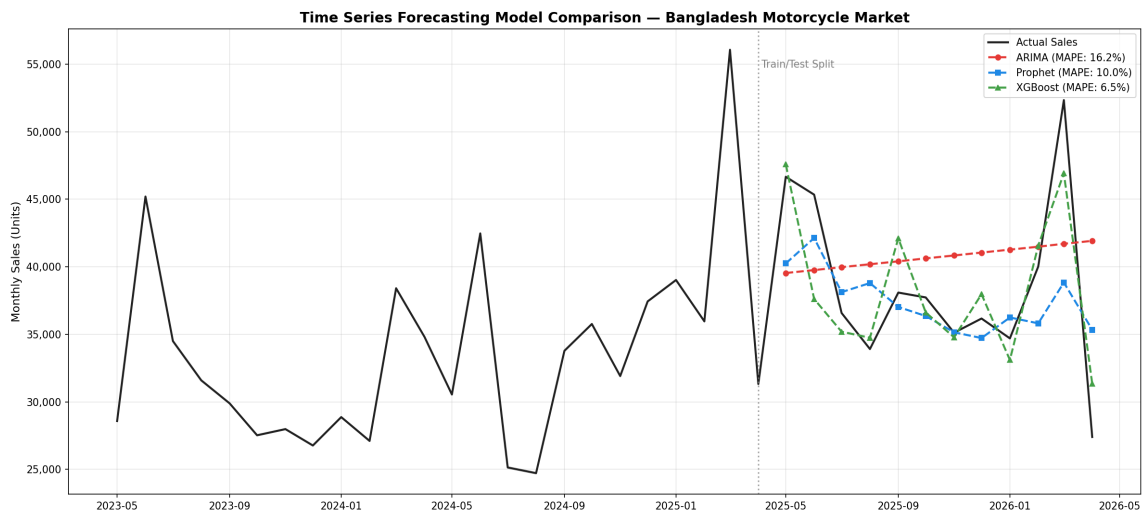


Figure 13: Time Series Forecasting Model Comparison — Actual vs Predicted

6.4 XGBoost Feature Importance

The XGBoost feature importance analysis reveals which factors are most predictive of monthly demand levels. The 12-month lag (sales from the same month in the previous year) emerges as the single most important feature, confirming the strong annual seasonality in motorcycle demand. The 1-month and 2-month lags are also highly ranked, indicating that recent demand momentum is a strong predictor of near-term sales. The 12-month rolling mean captures the underlying trend level, while month and quarter features encode the seasonal pattern. These findings validate the commercial intuition that

motorcycle demand is heavily influenced by seasonal cycles, recent market momentum, and the baseline demand level established over the preceding year.

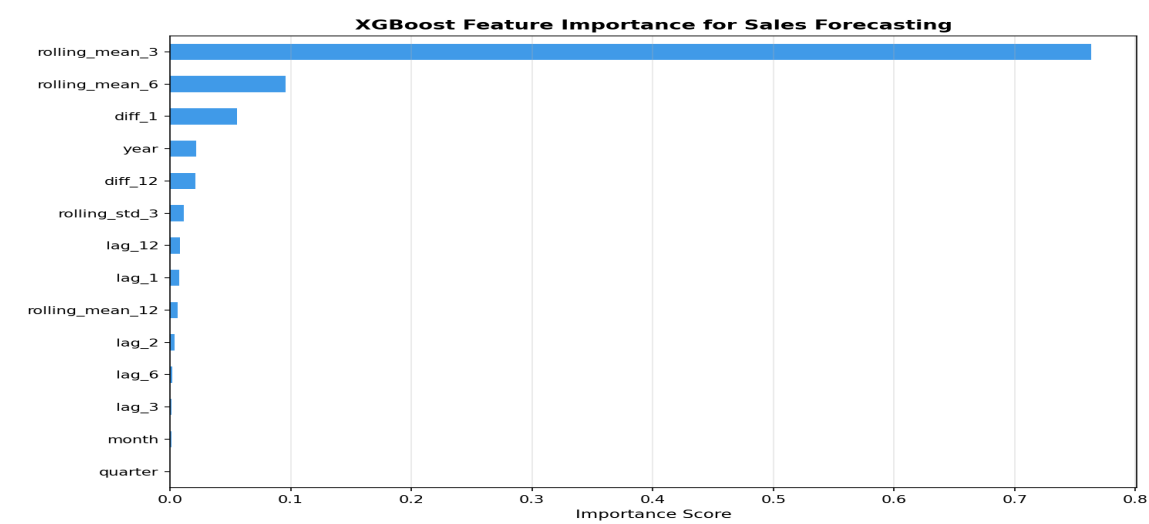


Figure 14: XGBoost Feature Importance for Sales Forecasting

6.5 Extended 12-Month Demand Forecast

Using the Prophet model with full historical data through April 2026, we generated a 12-month demand forecast for the Bangladesh motorcycle wholesale market. The forecast projects continued recovery momentum, with monthly wholesale volumes expected to average between 35,000 and 50,000 units over the forecast horizon. The seasonal pattern is expected to persist, with March-May showing the strongest demand and November-December showing seasonal softness. The 95% confidence interval widens over the forecast horizon, reflecting increasing uncertainty, but the central tendency suggests the market is on a recovery trajectory that could see annual volumes return to the 450,000-500,000 unit range in the medium term.

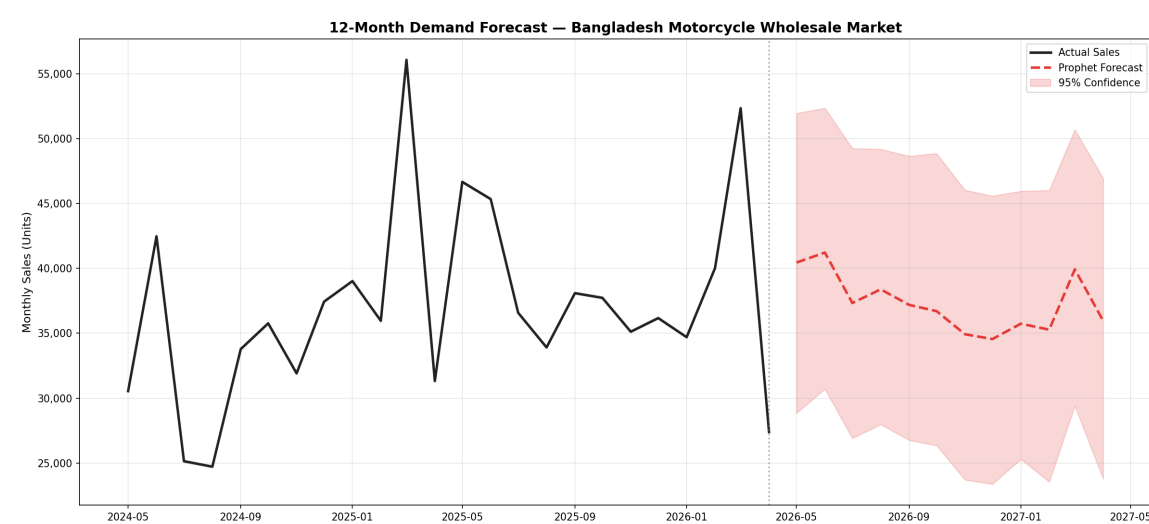


Figure 15: 12-Month Extended Demand Forecast with 95% Confidence Interval

6.6 Seasonal Decomposition

The additive seasonal decomposition separates the total market series into trend, seasonal, and residual components. The trend component clearly shows the three-phase market lifecycle: steady growth (2012-2016), explosive expansion (2017-2021), and contraction-stabilization (2022-2026). The seasonal component confirms the consistent annual pattern with March-June peaks and November-December troughs. The residual component captures irregular events including the COVID-19 disruption, policy shocks, and supply chain disruptions. The relatively small magnitude of residuals compared to the trend and seasonal components indicates that the market is predominantly driven by structural and cyclical factors rather than random shocks.

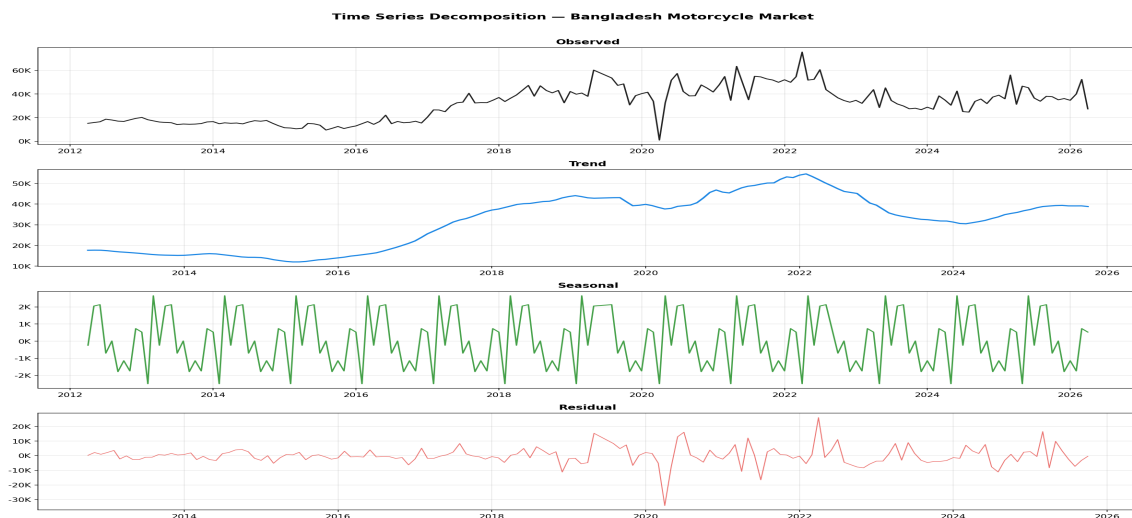


Figure 16: Additive Seasonal Decomposition of Total Market Sales

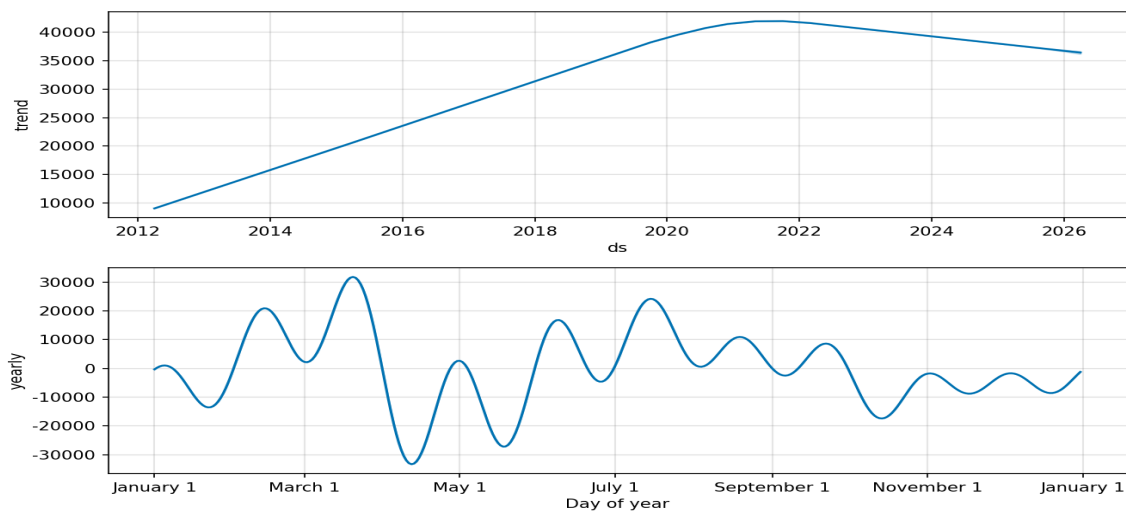


Figure 17: Prophet Model Decomposition — Trend, Yearly Seasonality, and Holiday Components

6.7 Brand-Level Forecasting

Individual brand-level forecasts were generated using the Prophet model for the five major brands. Honda and Yamaha are projected to maintain their growth trajectories, with Honda potentially reaching 8,000-9,000 units per month and Yamaha maintaining 7,000-8,000 units. Suzuki's strong recent momentum is expected to continue, potentially consolidating its position as the third major player. Bajaj is forecast to stabilize at approximately 5,000-6,000 units monthly, while Hero is projected to remain in the 5,000-7,000 unit range with moderate growth. These brand-level forecasts should be used as directional indicators rather than precise predictions, as brand-level dynamics are more susceptible to product launch timing, competitive moves, and supply chain disruptions.

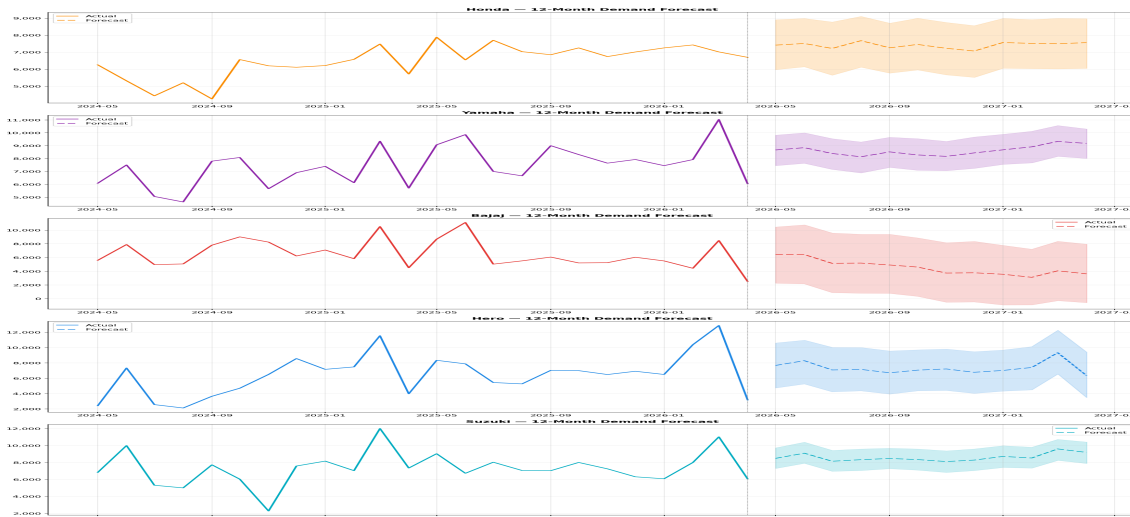


Figure 18: Brand-Level 12-Month Demand Forecasts (Honda, Yamaha, Bajaj, Hero, Suzuki)

7. Correlation and Pattern Mining

7.1 Brand Sales Correlation Analysis

The brand sales correlation matrix reveals important competitive dynamics and market structure relationships. Honda and Yamaha show a strong positive correlation (0.88), indicating that these brands tend to move together with the overall market cycle, likely because they share similar target segments and respond to the same demand drivers. Bajaj shows a notable negative correlation with Honda (-0.41) and Yamaha (-0.38), suggesting competitive displacement effects where gains by Honda and Yamaha come partly at Bajaj's expense. TVS shows weak positive correlation with most brands, reflecting its declining and increasingly idiosyncratic market position.

The correlation between Runner and other brands is generally weak, which is consistent with Runner's position in the entry-level segment where demand is driven by different factors (price sensitivity, rural connectivity) than the mid-range and premium segments where the major brands compete. These correlation patterns have practical implications for portfolio management: brands with low or negative correlation can provide diversification benefits for multi-brand distributors, while highly correlated brands may amplify inventory risk during market downturns.

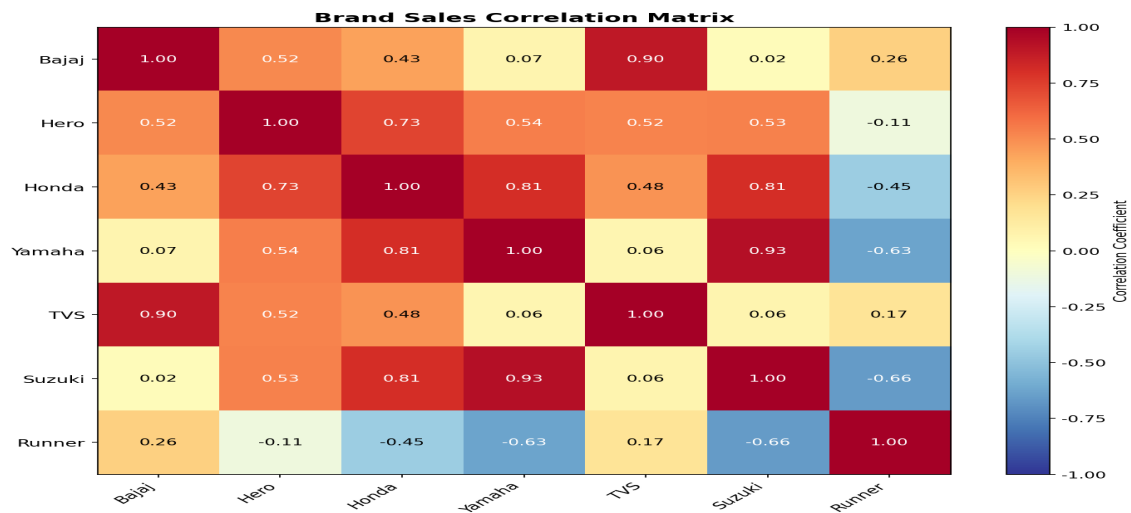


Figure 19: Brand Sales Correlation Matrix

7.2 Market Leadership Transitions

The pattern mining analysis identified 19 market leadership transitions over the 14-year period, with the frequency of transitions increasing dramatically in recent years. In the early period (2012-2020), Bajaj held market leadership almost continuously with only brief interruptions. However, from 2021 onward, leadership has rotated among Honda, Yamaha, Suzuki, Hero, and Bajaj with increasing frequency, reflecting the market's transition from monopoly to competitive oligopoly. The most recent transitions show monthly leadership changes, indicating that no brand has established sustainable competitive advantage in the current market environment.

The pattern mining also revealed that the longest consecutive growth streak is 4 months, while the longest decline streak is 6 months, confirming that the market is more prone to sustained declines than sustained growth phases. The top growth periods are all associated with recovery from disruptions: the post-COVID rebound (July 2020: +5,424% over 3 months), the 2021 recovery surge (May 2021: +83%), and the 2017 market expansion (August 2017: +62% over 4 months). These patterns suggest that the most significant demand creation opportunities come from recovery periods following market disruptions, rather than from organic growth trends.

8. Strategic Recommendations

8.1 Demand Forecasting and Inventory Management

Based on the forecasting model comparison and seasonal pattern analysis, we recommend that industry participants adopt a hybrid forecasting approach that combines the XGBoost model for short-term (1-3 month) predictions with the Prophet model for medium-term (3-12 month) strategic planning. The XGBoost model's superior accuracy (6.5% MAPE) makes it the preferred tool for operational decisions such as monthly production scheduling and dealer allocation, while Prophet's interpretable decomposition and automatic holiday handling make it better suited for annual planning and budgeting.

cycles. Inventory build-up should begin 60-90 days before the Eid demand window (January-February for March-May peak), with a 15-20% safety stock buffer above the forecast baseline to accommodate demand upside risk.

8.2 Competitive Strategy

The competitive analysis reveals that the market is in a transitional phase where first-mover advantage in emerging segments can create lasting competitive positions. Brands should focus on three strategic priorities: (1) Strengthening presence in the 125-150cc segment, which is the fastest-growing and most profitable category; (2) Expanding dealer network coverage in tier-2 and tier-3 cities where demand growth is outpacing urban centers; and (3) Building brand loyalty through after-sales service and financing partnerships, which are becoming critical differentiators as product features commoditize across brands.

For Bajaj specifically, the analysis suggests the need for a fundamental strategic review. The brand's consistent market share decline indicates that its current product portfolio and positioning are not resonating with the evolving Bangladeshi consumer. A potential turnaround strategy could include re-entering the high-volume 100-110cc commuter segment with a competitively priced product, leveraging the KTM brand equity for the premium segment, and investing in dealer network revitalization with a focus on customer experience and digital engagement.

8.3 Seasonal Planning

The seasonal demand analysis provides clear operational guidelines for production and inventory planning. Manufacturers should plan for peak capacity utilization in January-March to build inventory ahead of the Eid demand window. Post-Eid (June-August), production should be moderated to prevent inventory buildup during the seasonal trough. The winter period (November-February) should be used for model year transitions, dealer training, and network expansion activities that are difficult to execute during the high-demand season. Marketing investment should be front-loaded into the January-March period to capture demand during the peak selling season, with reduced spend during the summer trough and a secondary push in September-October for the post-monsoon recovery.

8.4 Risk Mitigation

The anomaly detection results highlight the importance of building resilience against demand shocks. The COVID-19 experience demonstrated that wholesale demand can drop by over 97% in a single month, creating severe cash flow and inventory management challenges for the entire supply chain. We recommend that industry participants maintain a 90-day cash reserve, establish flexible supply agreements with component suppliers that allow volume adjustments within 30 days, and develop scenario-based contingency plans for demand disruptions exceeding 30% of baseline. The current macroeconomic environment, characterized by taka depreciation, rising import costs, and potential regulatory changes, suggests that the probability of significant demand shocks remains elevated, making risk mitigation a strategic imperative rather than an optional consideration.

9. Methodology

9.1 Data Mining Techniques

The analysis employed a multi-method data mining approach to ensure robust pattern discovery. K-Means clustering was used for unsupervised demand pattern segmentation, with the optimal cluster count determined by silhouette score evaluation across $k=2$ to $k=6$. Feature scaling was performed using StandardScaler to normalize the seven-dimensional feature space. The Isolation Forest algorithm was applied with a 10% contamination parameter, which was calibrated based on domain knowledge of the expected frequency of market disruptions (approximately 1-2 significant disruptions per year). The algorithm's anomaly scores were validated against known market events to ensure detection accuracy.

9.2 Forecasting Methodology

The forecasting methodology followed a rigorous train-test split protocol, using the first 154 months for training and the final 12 months for out-of-sample evaluation. For ARIMA, an exhaustive grid search over p (0-3), d (0-2), and q (0-3) parameter space was conducted, selecting the model with the lowest Akaike Information Criterion (AIC). The Prophet model was configured with yearly seasonality enabled and a changepoint prior scale of 0.05 to balance trend flexibility with overfitting prevention. The XGBoost model used 200 estimators with a maximum depth of 5 and learning rate of 0.05, with features including temporal variables, lagged sales values (1, 2, 3, 6, 12 months), rolling statistics (3-month and 12-month means and standard deviations), and differenced series. Model performance was evaluated using MAPE and RMSE on the held-out test set.

10. Conclusion

This machine learning-driven analysis of the Bangladesh motorcycle wholesale market reveals a market in transition, characterized by shifting competitive dynamics, pronounced seasonal patterns, and significant structural breaks driven by external shocks. The key findings can be summarized as follows: the market has shifted from Bajaj monopoly to competitive oligopoly, with Honda, Yamaha, and Suzuki emerging as the new triumvirate; seasonal patterns are strongly tied to Eid festivals and monsoon cycles, with March-June representing the critical demand window; the XGBoost model provides the most accurate short-term demand forecasts, while Prophet offers the best interpretable medium-term projections; and anomaly detection confirms that the COVID-19 disruption was the most extreme demand shock in the market's history, but the market has demonstrated strong recovery capacity.

The practical implications of this analysis are significant for all market participants. Manufacturers should align production capacity and inventory planning with the identified seasonal patterns, ensuring adequate stock availability during the Eid demand window while avoiding overstock during seasonal troughs. Distributors should diversify their brand portfolios to mitigate concentration risk, particularly as the market becomes more competitive and brand shares continue to shift. Dealers should invest in digital capabilities and customer relationship management to capture demand more efficiently in an

increasingly competitive retail environment. The analytical framework presented in this report provides a foundation for ongoing market monitoring and strategic planning, and should be updated regularly as new data becomes available to capture evolving market dynamics.